Edugenie: Google Gemini Powered Learning Assistant

Vetri Thiran Payirchi Thittam

Topic: Module Implementation

1. Introduction

Module implementation is the stage where the Edugenie learning assistant is divided into functional modules and each module is developed, connected, tested, and integrated into the complete learning application. Edugenie uses Google Gemini to provide an interactive and personalized learning experience for students.

2. Main Modules of Edugenie

Module	Purpose
User Interface Module	Provides the web interface for students to enter questions and view responses.
Authentication Module	Manages user login and basic user access.
Question Processing Module	Receives student questions and prepares them for the AI model.
Gemini AI Module	Sends prompts to Google Gemini and receives AI-generated educational responses.
Learning Assistance Module	Provides explanations, examples, summaries, and study support.
Response Display Module	Displays Gemini responses in a clear and readable format.
Data/History Module	Can store learning interactions and useful user information when required.

3. Module Implementation Steps

- **Step 1 – Design the interface:** Create pages for login, question input, response display, and learning resources.
- **Step 2 – Configure the application:** Set up the development environment, project structure, required libraries, and configuration files.
- **Step 3 – Implement question processing:** Capture the student's question and validate the input before sending it to the AI service.
- **Step 4 – Integrate Google Gemini:** Connect the application to the Gemini API and create suitable prompts for educational responses.
- **Step 5 – Implement response handling:** Receive the model response, handle errors, and present the result to the student.
- **Step 6 – Add learning features:** Implement features such as explanations, summaries, examples, and topic-based assistance.
- **Step 7 – Test each module:** Test modules individually and then perform integration testing.

- **Step 8 – Integrate the complete system:** Connect all modules and verify the complete learning workflow.

4. Basic System Flow

Student → Web Interface → Question Processing → Gemini AI → Response Processing → Learning Response → Student

5. Example Module Workflow

When a student enters a question, the User Interface Module collects the question. The Question Processing Module validates and prepares it as a prompt. The Gemini AI Module sends the prompt to Google Gemini. The generated response is received, checked for basic errors, and passed to the Response Display Module. The student can then read the explanation and continue learning.

6. Testing and Validation

- Check whether every module works independently.
- Verify that questions are correctly transferred to the Gemini service.
- Check that useful and understandable responses are displayed.
- Test invalid or empty inputs and API/network errors.
- Verify the complete application on the target device or browser.

7. Expected Outcome

After module implementation, Edugenie provides an integrated AI-powered learning assistant that can accept student questions, communicate with Google Gemini, and present educational responses through a simple interface. Modular development also makes the application easier to test, maintain, and enhance in the future.

8. Conclusion

The module implementation phase converts the Edugenie design into a working learning system. By separating the application into clear modules and integrating Google Gemini, the project can provide students with interactive and accessible learning assistance while maintaining a structured and scalable architecture.